

Assignment 2

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1 Exercise 1.8

If $\mu = 0.9$, what is the probability that a sample of 10 marbles will have $\nu \leq 0.1$?

Key Points:

- The number of trials $N = 10$ (sample size)
- The probability of success $\mu = 0.9$ (drawing a red marble)
- Looking for the probability of $\nu \leq 0.1$ (drawing 1 or no red marbles)

Approach: This problem can be modeled using the binomial distribution. The binomial distribution is used to find the probability of getting a certain number of successes in a fixed number of independent trials, where the probability of success in each trial is constant.

The probability mass function of the binomial distribution is given by:

$$P(X = k) = \binom{N}{k} \mu^k (1 - \mu)^{N-k} \quad (1)$$

where:

- $P(X = k)$ is the probability of getting exactly k successes (red marbles).
- $\binom{N}{k}$ is the number of ways to choose k successes out of N trials.
- μ^k is the probability of getting k red marbles (successes).
- $(1 - \mu)^{N-k}$ is the probability of getting $N - k$ green marbles (failures).

We are interested in the cumulative probability for $k = 0$ and $k = 1$, i.e., $P(X = 0) + P(X = 1)$.

Solution:

1.1 Step 1: Calculate $P(X = 0)$

For $k = 0$ (no red marbles), the probability is:

$$P(X = 0) = \binom{10}{0} \cdot (0.9)^0 \cdot (0.1)^{10}$$

$$P(X = 1) = 10 \cdot 0.9 \cdot (0.1)^9 = 9 \cdot 10^{-9}$$

1.2 Step 2: Calculate $P(X = 1)$

For $k = 1$ (one red marble), the probability is:

$$P(X = 1) = \binom{10}{1} \cdot (0.9)^1 \cdot (0.1)^9$$

$$P(X = 1) = 10 \cdot 0.9 \cdot (0.1)^9 = 10 \cdot 0.9 \cdot 10^{-9} = 9 \times 10^{-9}$$

1.3 Step 3: Calculate the cumulative probability

We now sum the probabilities for $k = 0$ and $k = 1$:

$$P(X \leq 1) = P(X = 0) + P(X = 1)$$

$$P(X \leq 1) = 10^{-10} + 9 \times 10^{-9} = 9.1 \times 10^{-9}$$

1.4 Final Answer

The probability that a sample of 10 marbles will have $v \leq 0.1$ (i.e., 0 or 1 red marble) when $\mu = 0.9$ is approximately:

$$9.1 \times 10^{-9}$$

This is indeed a very small probability, indicating that such an outcome is extremely unlikely.

2 Exercise 1.9

Hoeffding Inequality

The Hoeffding Inequality is a probabilistic bound that measures how close the sample mean v is to the true mean μ . The inequality is given as:

$$P(|v - \mu| \geq \epsilon) \leq 2 \exp(-2N\epsilon^2)$$

Where:

- v is the sample mean (fraction of red marbles).
- $\mu = 0.9$ is the true mean (probability of drawing a red marble).
- ϵ is the deviation from the true mean μ (how far away v is from μ).
- $N = 10$ is the number of trials (sample size).

We are interested in the probability that $v \leq 0.1$, which corresponds to:

$$|v - \mu| \geq |\mu - v| = |0.9 - 0.1| = 0.8$$

Thus, $\epsilon = 0.8$.

2.1 Step 1: Plug values into Hoeffding Inequality

$$P(|v - \mu| \geq 0.8) \leq 2 \exp(-2 \times 10 \times (0.8)^2)$$

2.2 Step 2: Perform the calculations

$$P(|v - \mu| \geq 0.8) \leq 2 \exp(-2 \times 10 \times 0.64)$$

$$P(|v - \mu| \geq 0.8) \leq 2 \exp(-12.8)$$

$$P(|v - \mu| \geq 0.8) \leq 2 \times \exp(-12.8)$$

$$P(|v - \mu| \geq 0.8) \leq 2 \times 2.75 \times 10^{-6}$$

$$P(|v - \mu| \geq 0.8) \leq 5.5 \times 10^{-6}$$

2.3 Step 3: Compare with the result from Exercise 1.8

In Exercise 1.8, we computed the exact probability of observing 0 or 1 red marble (corresponding to $v \leq 0.1$) using the binomial distribution, and the result was approximately 9.1×10^{-9} .

Here, using the Hoeffding Inequality, we derived an upper bound on this probability: 5.5×10^{-6} . The Hoeffding bound is much larger than the exact probability calculated using the binomial distribution because it provides a loose bound, ensuring that the probability does not exceed this value but doesn't attempt to compute the exact probability.

2.4 Conclusion

Using the Hoeffding Inequality, we get an upper bound of 5.5×10^{-6} for the probability that $v \leq 0.1$, whereas the exact probability calculated previously was 9.1×10^{-9} . The Hoeffding bound is looser, but guarantees that the probability does not exceed this value.

3 Exercise 1.10

We flipped 1,000 coins 10 times and focused on three coins: - ν_1 : Fraction of heads for the first coin. - ν_{rand} : Fraction of heads for a randomly chosen coin. - ν_{min} : Fraction of heads for the coin with the minimum number of heads.

3.1 Part (a): What is μ for the three coins selected?

For all three coins, μ represents the probability of heads for a fair coin, which is 0.5. Since all coins are fair:

$$\mu = 0.5$$

3.2 Part (b): Analyze the distributions of ν_1 , ν_{rand} , and ν_{min}

The histograms generated from the simulation reveal the following:

1. ν_1 (First Coin): - The distribution for the first coin is centered around

0.5, with most values ranging between 0.4 and 0.6. This is expected for a fair coin flipped 10 times. - Most of the values are concentrated around 0.5, showing typical behavior for a fair coin.

2. ν_{rand} (Random Coin): - The random coin behaves similarly to ν_1 , with its distribution also centered around 0.5. This is consistent with the behavior of randomly selecting a fair coin from the population of 1,000 coins.

3. ν_{min} (Coin with Minimum Heads): - The distribution of ν_{min} is heavily skewed toward values close to 0, indicating that this coin often had very few heads. This is expected because ν_{min} was specifically chosen to have the minimum number of heads across the 1,000 coins.

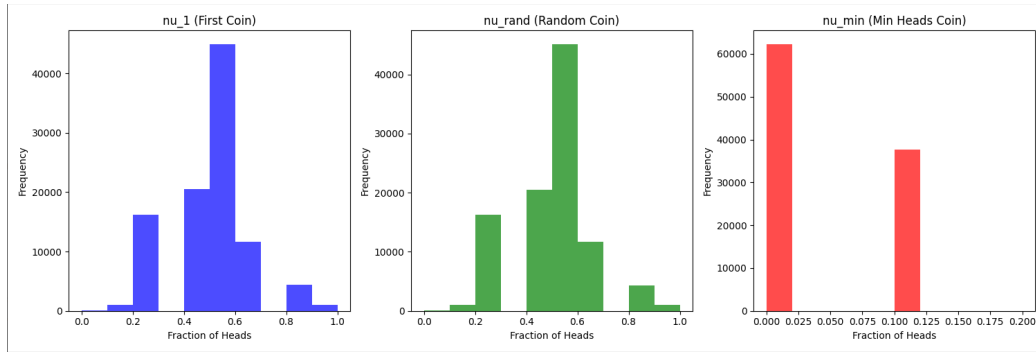


Figure 1: Generated Graphs by Python File

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3.3 Part (c): Compare estimates with the Hoeffding bound

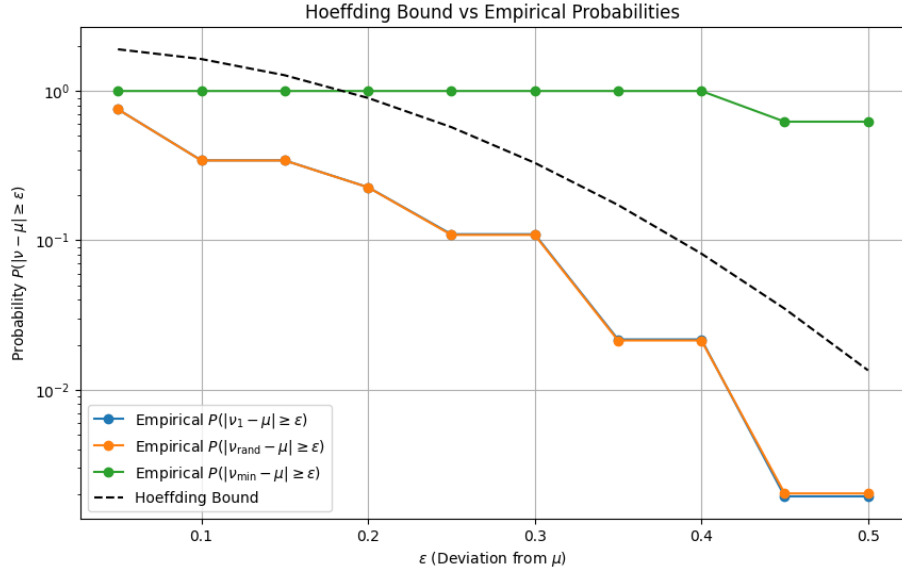


Figure 2: Generated Graphs by Python File

3.4 Part (d): Which coins obey the Hoeffding bound?

- ν_1 and ν_{rand} obey the Hoeffding bound because they represent specific or randomly selected coins. Their fractions of heads generally stay close to 0.5, with deviations consistent with the Hoeffding inequality.
- ν_{min} does not obey the Hoeffding bound. This is because the coin was chosen based on the extreme outcome (minimum number of heads), introducing a selection bias.

3.5 Part (e): Relating to multiple bins in Figure 1.10

The selection of $\nu_{\min}$ illustrates the concept of multiple bins, where if you have many hypotheses (or bins), one of them will naturally exhibit an extreme result by chance. In this case, $\nu_{\min}$ represents the extreme outcome of the coin with the least heads across the 1,000 trials.

This situation is similar to the concept of multiple hypothesis testing, where focusing on the most extreme outcome (in this case, the coin with the fewest heads) can lead to results that are not typical or representative of the general population of coins. This is why $\nu_{\min}$ does not follow the Hoeffding bound.

4 Exercise 1.11

4.1 Part (a)

No, S cannot guarantee better performance than random on points outside $\mathcal{D}$. While S selects the hypothesis that agrees most with the data in $\mathcal{D}$, there is no guarantee that this hypothesis will perform better than random guessing on points outside $\mathcal{D}$, since the target distribution outside $\mathcal{D}$ is unknown. The performance depends on the true label distribution of points outside $\mathcal{D}$.

4.2 Part (b)

Yes, it is possible. If all the examples in $\mathcal{D}$ have $y_n = +1$, S will choose h_1 , the hypothesis that predicts +1, while C will choose h_2 , the hypothesis that predicts -1 . However, if outside $\mathcal{D}$, the probability of $f(x) = +1$ is less than 0.5, then h_2 could perform better than h_1 on unseen points. Thus, it is possible that C 's hypothesis could outperform S 's hypothesis on points outside $\mathcal{D}$, depending on the true distribution of labels outside $\mathcal{D}$.

4.3 Part (c)

If $p = 0.9$, the probability that a randomly chosen point from the input space has label +1 is 0.9. Since S chooses the hypothesis that fits the data in $\mathcal{D}$, it will likely select h_1 , which predicts +1. C will choose the opposite hypothesis, h_2 , which predicts -1 .

On any new point, the probability that $f(x) = +1$ is 0.9, so S 's hypothesis h_1 will be correct with probability 0.9, while C 's hypothesis h_2 will be correct with probability 0.1.

Thus, the probability that S produces a better hypothesis than C is 0.9.

4.4 Part (d)

Yes, if $p < 0.5$, it is more likely that C will produce a better hypothesis than S . In this case, the probability of $f(x) = +1$ is less than the probability of $f(x) = -1$. S will still select h_1 (because $\mathcal{D}$ likely has more examples with $y = +1$), but the target function $f(x)$ is more likely to be -1 outside $\mathcal{D}$. Therefore, C 's hypothesis h_2 , which predicts -1 , would be more likely to perform better on new data points than S 's hypothesis.

5 Exercise 1.12 Solution

The best promise you can make is option (b):

After learning, you will provide her with a hypothesis g , and with high probability, the g which you produce will approximate f well out of sample.

5.1 Analysis

This promise is the most realistic and achievable given the situation. Since the target function f is completely unknown, and the learning process is inherently probabilistic, it is not possible to offer a **guarantee** (as option (a) suggests) that the hypothesis g will approximate f well out of sample.

However, with a sufficient amount of data (4,000 data points), it is likely that the hypothesis g will generalize well to unseen data. In machine learning, we typically provide guarantees of performance with high probability but not absolute certainty. Therefore, option (b) aligns with the nature of machine learning and the available data, making it the best promise you can offer in this scenario.

6 Problem 1.3 Solution

We are asked to prove that the Perceptron Learning Algorithm (PLA) eventually converges to a linear separator for separable data.

Let:

- $\mathbf{w}^*$ be an optimal set of weights that correctly separates the data.
- $\mathbf{w}(t)$ be the weight vector at iteration t of the Perceptron Learning Algorithm.

We are also given that $\mathbf{w}(0) = 0$ for simplicity.

6.1 Step (a)

Let $\rho = \min_{1 \leq n \leq N} y_n(\mathbf{w}^* \cdot \mathbf{x}_n)$. We need to show that $\rho > 0$.

Since the data is separable, there exists a hyperplane defined by $\mathbf{w}^*$ that perfectly separates the data. Therefore, for each point $\mathbf{x}_n$, the signed distance from $\mathbf{x}_n$ to the separating hyperplane is $y_n(\mathbf{w}^* \cdot \mathbf{x}_n)$.

Since no point lies exactly on the hyperplane, the signed distance is strictly positive, meaning that $\rho > 0$.

6.2 Step (b)

We now show that:

$$\mathbf{w}(t) \cdot \mathbf{w}^* \geq \mathbf{w}(t-1) \cdot \mathbf{w}^* + \rho$$

and conclude that:

$$\mathbf{w}(t) \cdot \mathbf{w}^* \geq t\rho.$$

The weight update at iteration $t-1$ is:

$$\mathbf{w}(t) = \mathbf{w}(t-1) + y_{t-1}\mathbf{x}_{t-1}.$$

Taking the dot product with $\mathbf{w}^*$, we get:

$$\mathbf{w}(t) \cdot \mathbf{w}^* = (\mathbf{w}(t-1) + y_{t-1}\mathbf{x}_{t-1}) \cdot \mathbf{w}^*.$$

Since $y_{t-1}(\mathbf{w}(t-1) \cdot \mathbf{x}_{t-1}) \leq 0$ (because the point was misclassified), we have:

$$\mathbf{w}(t) \cdot \mathbf{w}^* = \mathbf{w}(t-1) \cdot \mathbf{w}^* + y_{t-1}(\mathbf{x}_{t-1} \cdot \mathbf{w}^*).$$

By the definition of ρ , we know that $y_{t-1}(\mathbf{x}_{t-1} \cdot \mathbf{w}^*) \geq \rho$, so:

$$\mathbf{w}(t) \cdot \mathbf{w}^* \geq \mathbf{w}(t-1) \cdot \mathbf{w}^* + \rho.$$

By induction, this leads to:

$$\mathbf{w}(t) \cdot \mathbf{w}^* \geq t\rho.$$

6.3 Step (c)

Next, we show that:

$$\|\mathbf{w}(t)\|^2 \leq \|\mathbf{w}(t-1)\|^2 + \|\mathbf{x}(t-1)\|^2.$$

Using the update rule:

$$\mathbf{w}(t) = \mathbf{w}(t-1) + y_{t-1}\mathbf{x}(t-1),$$

taking the squared norm of both sides, we get:

$$\|\mathbf{w}(t)\|^2 = \|\mathbf{w}(t-1) + y_{t-1}\mathbf{x}(t-1)\|^2.$$

Expanding the norm:

$$\|\mathbf{w}(t)\|^2 = \|\mathbf{w}(t-1)\|^2 + 2y_{t-1}(\mathbf{w}(t-1) \cdot \mathbf{x}(t-1)) + \|\mathbf{x}(t-1)\|^2.$$

Since $y_{t-1}(\mathbf{w}(t-1) \cdot \mathbf{x}(t-1)) \leq 0$ (as the point was misclassified), we have:

$$\|\mathbf{w}(t)\|^2 \leq \|\mathbf{w}(t-1)\|^2 + \|\mathbf{x}(t-1)\|^2.$$

6.4 Step (d)

We now show by induction that:

$$\|\mathbf{w}(t)\|^2 \leq tR^2,$$

where $R = \max_{1 \leq n \leq N} \|\mathbf{x}_n\|$.

From step (c), we have:

$$\|\mathbf{w}(t)\|^2 \leq \|\mathbf{w}(t-1)\|^2 + R^2.$$

By induction, we conclude:

$$\|\mathbf{w}(t)\|^2 \leq tR^2.$$

6.5 Step (e)

Using steps (b) and (d), we now show that:

$$\frac{\mathbf{w}(t) \cdot \mathbf{w}^*}{\|\mathbf{w}(t)\|} \geq \sqrt{t} \cdot \frac{\rho}{R},$$

and hence prove that:

$$t \leq \frac{R^2 \|\mathbf{w}^*\|^2}{\rho^2}.$$

From step (b), we have:

$$\mathbf{w}(t) \cdot \mathbf{w}^* \geq t\rho,$$

and from step (d), we have:

$$\|\mathbf{w}(t)\|^2 \leq tR^2.$$

Thus:

$$\frac{\mathbf{w}(t) \cdot \mathbf{w}^*}{\|\mathbf{w}(t)\|} \geq \frac{t\rho}{\sqrt{t}R} = \sqrt{t} \cdot \frac{\rho}{R}.$$

Since $\frac{\mathbf{w}(t) \cdot \mathbf{w}^*}{\|\mathbf{w}(t)\|} \leq \|\mathbf{w}^*\|$, we have:

$$\sqrt{t} \cdot \frac{\rho}{R} \leq \|\mathbf{w}^*\|.$$

Squaring both sides gives:

$$t \leq \frac{R^2 \|\mathbf{w}^*\|^2}{\rho^2}.$$

6.6 Conclusion

Thus, we have shown that the Perceptron Learning Algorithm converges in at most:

$$t \leq \frac{R^2 \|\mathbf{w}^*\|^2}{\rho^2}$$

iterations, provided the data is linearly separable.

7 Problem 1.7 Solution

We are given a binomial distribution for a sample of heads and tails generated by tossing a coin N times independently. The probability of obtaining k heads from N tosses for a given coin is given by:

$$P(k | N, \mu) = \binom{N}{k} \mu^k (1 - \mu)^{N-k}$$

Where: - μ is the probability of heads (probability of error), - N is the number of tosses, - k is the number of heads obtained.

The training error ν is given by:

$$\nu = \frac{k}{N}$$

7.1 Part (a)

We are asked to compute the probability that at least one coin will have $\nu = 0$ (i.e., no heads in all tosses) for different numbers of coins when $\mu = 0.05$ and $\mu = 0.8$. The cases to consider are 1 coin, 1,000 coins, and 1,000,000 coins.

The probability that a single coin has $\nu = 0$ (i.e., no heads) is:

$$P(\nu = 0 | N, \mu) = P(k = 0 | N, \mu) = (1 - \mu)^N$$

For $\mu = 0.05$, $N = 10$:

$$P(\nu = 0 | N = 10, \mu = 0.05) = (0.95)^{10} \approx 0.5987$$

For $\mu = 0.8$, $N = 10$:

$$P(\nu = 0 | N = 10, \mu = 0.8) = (0.2)^{10} = 1.024 \times 10^{-7}$$

The probability that no coin has $\nu = 0$ for M coins is:

$$P(\text{no coin has } \nu = 0) = (1 - P(\nu = 0))^M$$

Thus, the probability that at least one coin has $\nu = 0$ is:

$$P(\text{at least one coin has } \nu = 0) = 1 - (1 - P(\nu = 0))^M$$

Let's calculate this for the different values of M and μ .

- For $\mu = 0.05$:

1. $M = 1$:

$$P(\text{at least one coin has } \nu = 0) = 1 - (1 - 0.5987)^1 = 0.5987$$

2. $M = 1,000$:

$$P(\text{at least one coin has } \nu = 0) = 1 - (1 - 0.5987)^{1000} \approx 1$$

3. $M = 1,000,000$:

$$P(\text{at least one coin has } \nu = 0) = 1 - (1 - 0.5987)^{1000000} \approx 1$$

- For $\mu = 0.8$:

1. $M = 1$:

$$P(\text{at least one coin has } \nu = 0) = 1 - (1 - 1.024 \times 10^{-7})^1 \approx 1.024 \times 10^{-7}$$

2. $M = 1,000$:

$$P(\text{at least one coin has } \nu = 0) = 1 - (1 - 1.024 \times 10^{-7})^{1000} \approx 1.024 \times 10^{-4}$$

3. $M = 1,000,000$:

$$P(\text{at least one coin has } \nu = 0) = 1 - (1 - 1.024 \times 10^{-7})^{1000000} \approx 0.0952$$

7.2 Part (b)

We are asked to plot the probability:

$$P\left(\max_i |\nu_i - \mu_i| > \epsilon\right)$$

for $N = 6$ and 2 coins with $\mu_1 = \mu_2 = 0.5$, for $\epsilon \in [0, 1]$, and compare the result to the bound provided by the Hoeffding inequality. The Hoeffding bound for a single coin is:

$$P(|\nu - \mu| > \epsilon) \leq 2e^{-2N\epsilon^2}$$

The probability that the maximum deviation between ν_i and μ_i exceeds ϵ for two coins is:

$$P\left(\max_i |\nu_i - \mu_i| > \epsilon\right) = P(|\nu_1 - \mu_1| > \epsilon \cup |\nu_2 - \mu_2| > \epsilon)$$

By the inclusion-exclusion principle:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Since the two coins are independent, $P(A \cap B) = P(A)P(B)$, so:

$$P\left(\max_i |\nu_i - \mu_i| > \epsilon\right) = 2P(|\nu - \mu| > \epsilon) - P(|\nu_1 - \mu_1| > \epsilon)P(|\nu_2 - \mu_2| > \epsilon)$$

Using the Hoeffding inequality for a single coin:

$$P(|\nu - \mu| > \epsilon) \leq 2e^{-2N\epsilon^2}$$

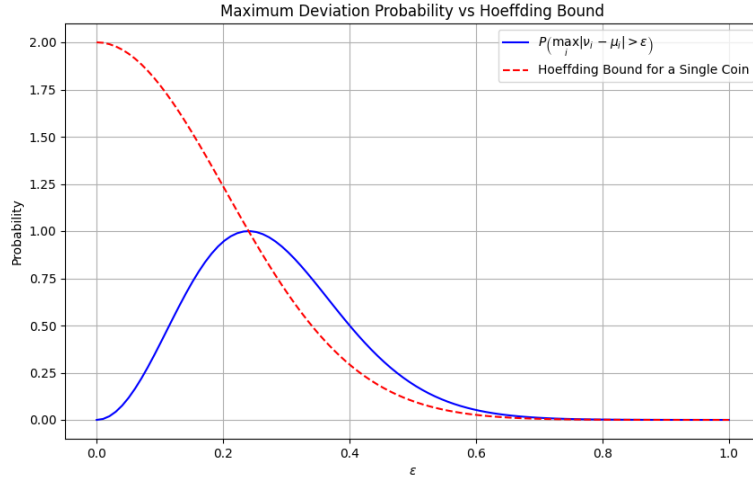


Figure 3: Results for $N = 6$ and epsilon in $[0,1]$